

CO3 AT2 - Question 2

Case Study: Live Sports Scoreboard Ranking (Quick Sort)

Aim:

To analyze the use of Quick Sort in dynamically ranking players and teams in a live sports analytics platform.

Problem Statement:

A live sports analytics platform continuously updates player and team rankings during ongoing matches. Analyze how Quick Sort can support dynamic sorting of scores in real time. Identify issues such as pivot selection, repeated updates, and worst-case behavior. Apply divide and conquer concepts and propose optimized pivot strategies.

Objectives:

- Implement Quick Sort for ranking scores.
- Analyze the effect of pivot selection.
- Study worst-case behavior and repeated updates.
- Propose optimized solutions for real-time systems.

Divide and Conquer Approach:

1. Select a pivot score.
2. Partition scores into lower and higher groups.
3. Recursively sort both groups.
4. Combine the results to obtain final rankings.

Issues Identified:

- Poor pivot selection may degrade performance.
- Continuous score updates require repeated sorting.
- Worst-case complexity $O(n^2)$ occurs for nearly sorted data.

Optimized Solution:

Use Randomized Pivot Selection or Median-of-Three Pivot Selection to produce balanced partitions and maintain near $O(n \log n)$ performance.

Why Quick Sort?

Quick Sort is fast in practice, cache-friendly, and suitable for dynamic ranking systems when enhanced with improved pivot strategies.

Time Complexity:

Best Case: $O(n \log n)$

Average Case: $O(n \log n)$

Worst Case: $O(n^2)$

Result and Analysis:

Quick Sort efficiently ranked live scores. Optimized pivot strategies reduced the probability of worst-case behavior and improved real-time performance.

Conclusion:

Quick Sort with randomized or median-of-three pivot selection is an effective choice for live sports ranking systems because it provides fast average-case performance and supports dynamic updates efficiently.

Output:

(Insert program output screenshot here.)